

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	41.90426	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	75.6328	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1379.29602	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	235.31786	1	0	0	0.972629	13.41
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	220.43222	1	0	0	0.923307	9.95
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	140.07968	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 41.90426.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 75.6328.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1379.29602.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 235.31786.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 220.43222.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 140.07968.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Aspect	Summary
Direct Synthesis	*direct_generate_plus_one_repair* failed to produce any feasible solutions (0% feasibility, gap penalty 3.0) despite low runtime and high novelty/complexity.
Budget-Matched Independent Search	*budget_matched_no_replay* also failed on feasibility (0%), matching direct synthesis in poor performance, despite slightly lower runtime.
Replay-Aware Iterative Search	*replay_solver_evolution* achieved 100% feasibility with feasible gap 0.273, but with no improvement over nearest neighbor baseline (gap 0.273) and slower runtime (140 ms vs 42 ms).
Fixed-Baseline Context	Clarke-Wright Savings was the best baseline with 100% feasibility, lowest feasible gap (0.074), and moderate runtime (76 ms), outperforming nearest neighbor (gap 0.273) and regret insertion local search (gap 0.466 but with much longer runtime ~1.38s).
Feasibility	All baselines and replay solver reached 100% feasibility; direct generation and budget-matched no replay failed completely.
Objective Gap	Clarke-Wright Savings baseline had the smallest objective gap (0.074), replay solver matched nearest neighbor gap (0.273), others worse.

Aspect	Summary
Runtime	Nearest neighbor fastest (42 ms), Clarke-Wright moderate (76 ms), replay slower (140 ms), regret insertion slowest (~1.38 s). Direct and budget-matched no replay were ~220-235 ms but infeasible.
Robustness Across Families	Clarke-Wright had the lowest robustness dispersion (0.0078), implying stable performance; others comparable or worse.
Learned Condition vs Baselines	No learned condition (direct synthesis, replay) clearly outperformed the best fixed baseline (Clarke-Wright Savings). Learned methods either matched weaker baseline or were infeasible.
Replay vs No-Replay Control	Replay-aware solver yielded feasible solutions with gaps similar to nearest neighbor but worse than Clarke-Wright, while no-replay control failed feasibility entirely. Replay was better than no-replay control but did not beat the best fixed baseline.

Conclusion

- ****The Clarke-Wright Savings baseline is the best performer, balancing feasibility, low gap, and reasonable runtime.****
- ****Learned direct synthesis and budget-matched no-replay methods failed feasibility.****
- ****Replay-assisted search improves feasibility over no-replay but does not surpass Clarke-Wright baseline performance.****
- ****No learned method clearly beats fixed baselines in objective gap or robustness.****